

Reversible Data Hiding for Medical-Image in VQ with Two-Layer Embedding

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Abstract: We present a reversible, two-layer data-hiding scheme for vector quantization images that operates on the index table rather than pixel values. A seed/residual block design with side-match evaluation determines embeddability without modifying content. For embeddable blocks, a 4-bit prefix enables the unique recovery of the original index. A compressed indication map for residual blocks is encoded using Extended Run-Length Encoding to limit signaling overhead. Under a 32-bit per-index budget for 4×4 blocks, the method separates address bits and payload while preserving codec compatibility and exact reversibility. Experiments on six standard medical images demonstrate high visual fidelity: the Peak Signal-to-Noise Ratio (PSNR) increases with codebook size K , while the embedding rate decreases mildly as the candidate sets become less homogeneous. Net payload is reported as effective capacity (EC). Comparisons of standard grayscale benchmarks indicate higher capacity than representative prior methods at $K = 64$, and competitive or superior results at $K = 256$. The approach offers a controllable capacity–fidelity trade-off suitable for high-integrity applications such as medical imaging.

I. INTRODUCTION

In today's telemedicine powered by IoT, large amounts of medical imagery and sensor data are collected at the edge and transmitted over various networks for clinical purposes. This raises longstanding concerns about confidentiality, integrity, and ownership. Compressed-domain data hiding that embeds critical metadata (e.g., patient identifiers, clinical notes, device telemetry) into the transmitted bitstream is therefore attractive, as it preserves diagnostic quality while streamlining end-to-end delivery [1][2][3].

Among the various methods of compression, VQ divides an image into fixed-size blocks and assigns each block to a codeword that has been learned. The compressed representation includes an index table and a codebook. The Linde–Buzo–Gray (LBG) algorithm is fundamental for designing the codebook, and ongoing improvements continue to enhance VQ's effectiveness in resource-constrained Internet of Things (IoT)

environments. Embedding at the index-table level is compelling because it aligns the payload with what is transmitted, enabling precise rate control with minimal added distortion after decoding [1].

VQ-domain hiding has progressed along several directions: reversible index coding for lossless host recovery [2], index-mapping to increase capacity with reversibility guarantees [3], side-match VQ (SMVQ)–based reversible embedding [4], two-layer designs combining de-clustering with indicator-free search-order coding (SOC) to camouflage modified indices [5], and lossless schemes that exploit adjacent-index correlation [6]. Collectively, these works demonstrate that reasoning over index-sequence correlation and coding paths yields substantial trade-offs between payload, quality, and rate [2][3][4][5][6]. In parallel, secure medical imaging emphasizes reversibility and robustness under bandwidth constraints. Compressed-domain strategies, such as Absolute Moment Block Truncation Coding (AMBTC) and IoT-oriented architectures, reduce overhead and attack surface

by fusing metadata with the image stream [7][8][9].

Two practical gaps remain for VQ index-table hiding in IoT workflows: (i) heterogeneous block content (smooth/ edge/ texture) makes a single embedding rule sub-optimal across anatomy and modality; and (ii) block-size sensitivity (e.g., 4×4 , 8×8 , 16×16) co-determines index correlation and codebook adjacency, shaping the achievable payload/bitrate/quality envelope, which refers to the balance between the amount of data that can be transmitted, the speed at which it can be transmitted, and the quality of the transmitted data.

We propose a privacy-preserving VQ index-table scheme tailored to IoT/telemedicine. The image is partitioned into VQ blocks; embedding is content- and role-adaptive (e.g., texture vs. ROI-proximal), and block size is selected to balance payload, bitrate, and decoded quality for the target deployment.

Contributions.

Blocks are classified by content and spatial role and assigned index-level rules that exploit correlation in complex areas while constraining operations near diagnostically sensitive boundaries. We sweep multiple VQ block sizes (4×4 , 8×8 , 16×16) and report trade-offs among payload, bitrate change, and PSNR/SSIM to guide IoT practitioners. Metadata is embedded directly in the index table to avoid duplicate transmissions, reducing latency and attack surface while remaining compatible with existing VQ encoders/decoders. The design supports near-lossless or reversible modes with low

side information and preserved visual fidelity, aligning with VQ-domain RDH [2][3][4][5][6]. Relative to index-mapping [3], SMVQ-based [4], de-clustering/SOC-based [5], and lossless prediction schemes [6], our method couples content-aware embedding with block-size selection and targets IoT-constrained medical pipelines end-to-end, preserving diagnostic visibility and scaling with device capabilities [3][4][5][6][7][8][9].

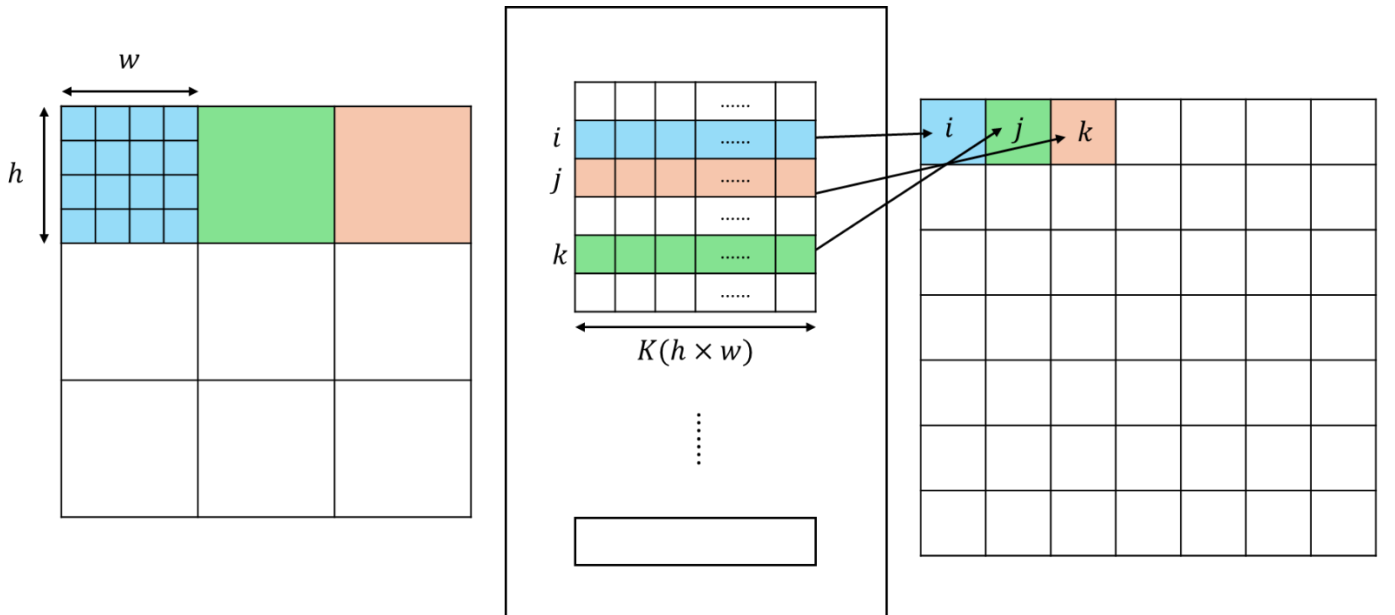
Organization.

Section II reviews VQ-domain hiding and IoT medical security. Section III details the adaptive embedding and block-size strategy. Section IV presents experiments across block sizes, payloads, and visual/bitrate metrics. Section V concludes with considerations for the deployment of telemedicine.

II. RELATED WORK

2.1 VQ Compression and Index-Table Data Hiding

Vector quantization (VQ) compresses images by dividing them into fixed-size blocks and mapping each block to a codeword in a trained codebook. The compressed stream consists of the codebook and an index table. The classic Linde–Buzo–Gray (LBG) algorithm remains the standard method for codebook design and supports most VQ-based embedding work [1].



◇ Fig. 1. Schematic of VQ image compression. Here, $(h \times w)$ denotes the block size; $(K = h \times w)$ is the codeword length (dimensionality) of the codebook; and (i, j, k) are indices for the image block position and the selected codeword, respectively.

Early reversible VQ-domain hiding directly targeted the index table. Chang et al. introduced locally adaptive coding (LAC) to free space in VQ indices while ensuring lossless recovery of the original host [2], followed by an improved reversible scheme for VQ indices using modified LAC [11]. Qin et al. proposed an index-mapping mechanism that increases capacity while remaining reversible [3]. Kieu and Chang utilized side-match VQ (SMVQ) and joint neighboring coding to embed data into indices with reversibility [4]. Parallel research explored search-order coding (SOC) paths and state-codebook mappings for reversible embedding in two-stage VQ (TSVQ) streams [12][13][14]. More recently, Chang et al. combined de-clustering with indicator-free SOC (IFSOC) to develop a two-layer reversible scheme that hides larger payloads in IFSOC-encoded index tables at relatively low bit-rate overhead [5]. Complementing these efforts, lossless VQ-domain designs utilize adjacent-index correlation (e.g., adaptive prediction-difference coding with codebook rearrangement) to compress index differences, thereby freeing capacity without degrading the decoded image [6]. Collectively, these works demonstrate that modeling index-sequence correlation and coding structure (LAC, SMVQ, SOC/IFSOC, TSVQ) is a practical approach to achieving payload capacity, reversibility, and rate control in the VQ compressed domain.

Two practical issues motivate our work. First, block heterogeneity (such as smooth, edge-rich, or textured regions) is rarely utilized explicitly in index-table embedding rules; most methods treat indices uniformly, even though local content influences index predictability and codebook adjacency. Second, the choice of block size (e.g., 4×4 , 8×8 , 16×16) is often fixed, yet it jointly determines index correlation and the trade-off between payload, quality, and bit-rate. Our approach addresses both by integrating content and region-aware index-table embedding with a block-size sweep for application-dependent selection.

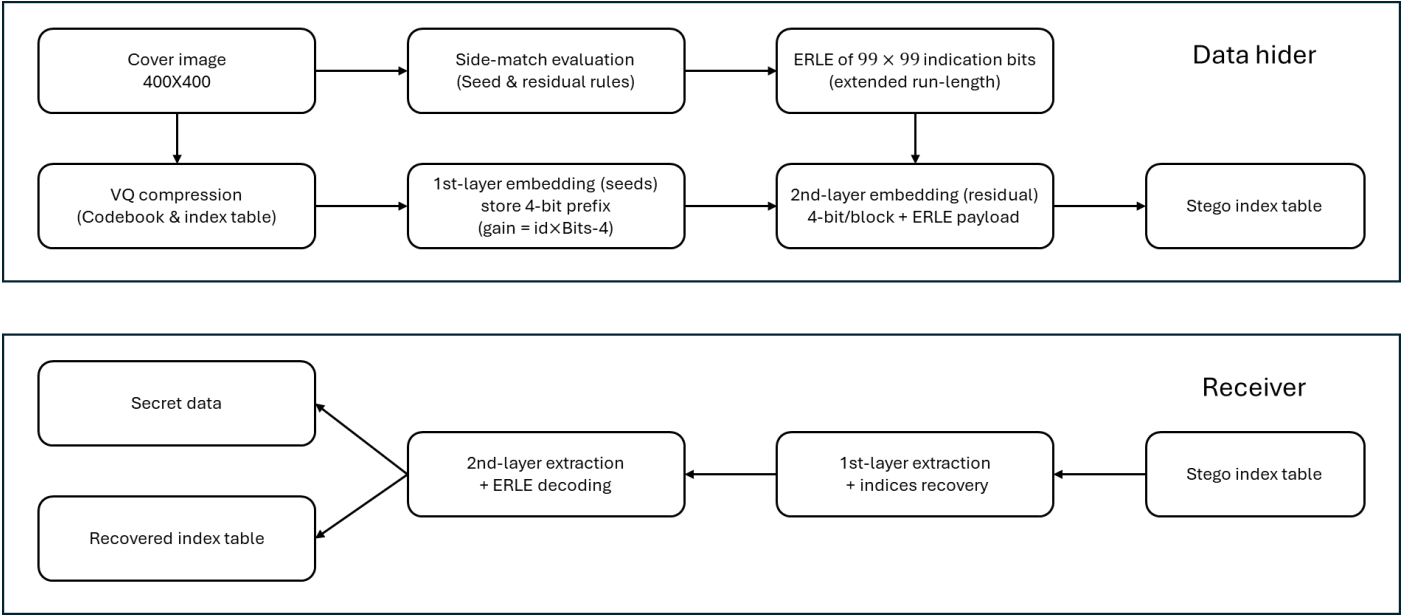
2.2 Compressed-Domain Hiding for IoT/Telemedicine

Telemedicine and IoT (or IoMT) pipelines continuously transmit diagnostic images and metadata over diverse networks; security surveys emphasize the importance of confidentiality, integrity, and reversibility, as well as ROI awareness, for clinical safety [7]. Embedding clinical metadata within the compressed stream (instead of sending it through a separate

channel) reduces latency. It minimizes the attack surface—an approach reflected in numerous AMBTC-domain studies, which, although not VQ-based, highlight the advantages of compressed-domain hiding in medical settings. For example, Ou and Sun classify AMBTC blocks as smooth or complex and embed data using bit planes or level ordering accordingly [15]. Kumar et al. incorporate a preprocessing filter and utilize Hamming distance and pixel-value differencing across three block types to increase capacity [16]. Subsequent work employs matrix-encoding/Hamming strategies, selective XOR, and two-layer procedures to further boost payload capacity with controlled distortion [17][18][19]. Recent studies focused on medical imaging (e.g., HPDH-MI) explicitly aim to achieve high payloads in AMBTC streams for hospital workflows [9]. In contrast, gradient-based AMBTC approaches align with eHealth/IoT deployment constraints and emphasize maintaining diagnostic quality [20]. These studies support our system-level assertion: embedding within the compressed domain benefits IoT/telemedicine by enhancing throughput and security, provided that diagnostic fidelity and reversibility are maintained [7], [9], [20]. Positioning. Compared to VQ-index RDH lines [2] [3] [4] [5] [6] [12] [13] [14], our method adapts explicitly index-table embedding to block roles (e.g., ROI-proximal vs. background) and empirically selects block sizes suited for the target IoT scenario. In relation to AMBTC-based medical pipelines [15] [16] [17] [18] [19] [20], our contribution preserves the advantages of compressed-domain embedding. Still, it advances them by incorporating them into the VQ index table, which naturally serves as the carrier for VQ-compressed imaging at the edge and in the cloud.

III. PROPOSED SCHEME

We propose a three-stage data-hiding scheme for VQ index tables. First, in the preprocessing stage, the input image is vector-quantized: the image is partitioned into blocks of the desired size, and a codebook is constructed. Second, in the first-layer embedding, payload bits are embedded into seed blocks; a side-match evaluation is employed to determine whether each block is embeddable. Third, in the second-layer embedding, payload bits are embedded into residual blocks judged



✧ Fig. 2. Flowchart of the proposed scheme.

embeddable by appending secret bits to the VQ index table so that the modified entries remain indistinguishable from a conventional VQ index table. The complete workflow, encompassing both the Data Hider and Receiver pipelines, is illustrated in Fig. 2. The following subsections provide detailed information on the preprocessing, first-layer embedding, and second-layer embedding procedures.

A. VQ Compression

We allocate 32 bits per index to carry both the compression index and hidden payload. The source image is partitioned into 4×4 blocks; each block is represented by a 32-bit entry that stores the VQ code and the embedded message bits. For a 400×400 image, this yields 100×100 blocks, so the index table has size $1 \times 10,000$, with each entry pointing to a codeword in the codebook.

After VQ training, the codebook has dimension $16 \times K$ (each codeword is 16-dimensional, corresponding to a 4×4 block flattened in row-major order). We consider four codebook sizes, $K \in \{64, 128, 256, 512\}$. The number of address bits required per index B is given by Eq. (1):

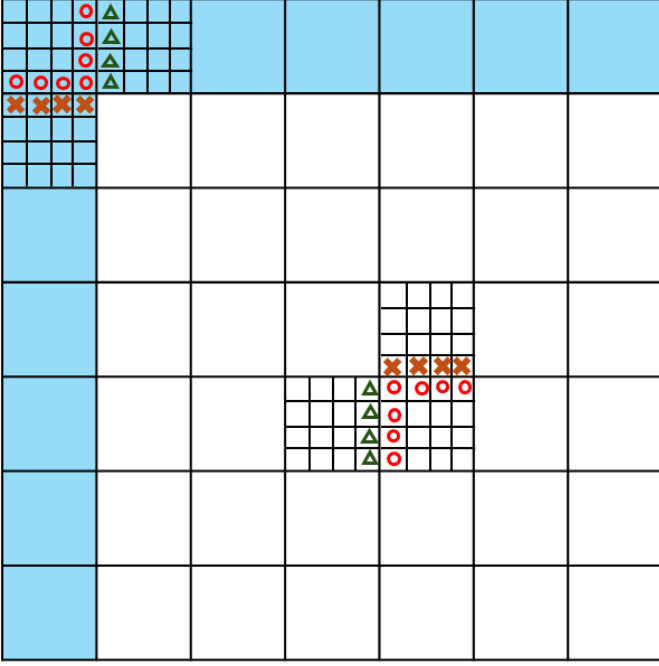
$$\text{indexBits} = B = \lceil \log_2 K \rceil \quad (1)$$

Thus, $B \in \{6, 7, 8, 9\}$ for the above K . The remaining bits (out of the 32-bit index) are available to encode auxiliary flags and/or payload, as specified in the embedding design.

B. Side-Match Evaluation

Below, we determine whether seed and residual blocks are embeddable, without modifying any pixels, by comparing a target pixel subvector from the block to a restricted subvector of candidate codewords. For each block, let the original index select a codeword; given a predefined candidate set (on dimensions chosen), we compute the Euclidean distance between the block's target subvector and each candidate's corresponding subvector. If the distance-minimizing candidate coincides with the original index, the block is embeddable as $\text{indication} = 0$; otherwise, it is non-embeddable as $\text{indication} = 1$. Concretely, we taking a 400×400 image with a block size of 4×4 as an example: top-row seeds compare only dimensions $\{1, 5, 9, 13\}$ —the first pixel in each row of the 4×4 block, highlighted by the green triangle in Fig. 3—against a target subvector drawn from positions $\{4, 8, 12, 16\}$ (the fourth pixel in each row, red circle in Fig. 3); left-column seeds compare only dimensions $\{1, 2, 3, 4\}$ (top row, orange symbols in Fig. 3) against targets from $\{13, 14, 15, 16\}$ (bottom row, red circle); and residual blocks compare dimensions $\{1, 2, 3, 4, 5, 9, 13\}$ (top row plus the first column's remaining three pixels; 7 total, red circle) against the block's own pixels at those positions. Figure 3 visualizes this seed/residual classification and yields a 100×100 indication map, where 0 = embeddable and 1 = non-embeddable. We apply ERLE only to the lower 99×99 residual region, since these 9,801 bits constitute the actual

ancillary overhead that must be transmitted or stored.



◇ Fig. 3. Schematic of the two-layer data-embedding framework.

C. First-Layer Data Embedding

For the 199 seed blocks (top row U left column, counting the top-left block once), let B be the index address length in bits and let L denote the stored prefix length; here $L = 4$ bits, the encoder stores only a L -bit prefix that enables unique recovery under the same side-match rule. An embeddable seed (indication = 0) frees $B - L$ bits for payload; a non-embeddable seed (indication = 1) contributes zero payload. Denoting by N_s the number of embeddable seed blocks and by P_s the total seed-layer payload, it is given by Eq. (2):

$$P_s = N_s(B - L) \quad (2)$$

where B is the index-bit length per address, L is the stored prefix length, N_s is the number of embeddable seed blocks, and P_s is the seed-layer payload.

D. Second-Layer Data Embedding

For residual blocks, an embeddable block stores only a L -bit prefix and contributes a fixed payload of L bits, while a non-embeddable block contributes zero; denoting by N_r the number of embeddable residual blocks and by P_r the total residual-layer payload, it is given by Eq. (3):

$$P_r = N_r L \quad (3)$$

where B is the index-bit length per address, L is the stored prefix length, N_r is the number of embeddable residual blocks, and P_r is the residual-layer payload.

E. Indication-Bit Overhead and ERLE

To signal embeddability in the residual region, we form a binary indication map over the 99×99 residual blocks, flatten it to a length-9801 sequence, and compress it with ERLE to obtain the bit length E_{RLE} . With P_s and P_r defined previously, the net payload is given by Eq. (4):

$$EC = P_s + P_r - E_{RLE}. \quad (4)$$

The file size in bits is given by Eq. (5):

$$F = \frac{I_h I_w}{b_h b_w} B + K(b_h b_w) q \quad (5)$$

where I is the image with height I_h and width I_w , $b_h b_w$ is the VQ block size, and q is the codebook sample precision so that each of the K codewords contains $b_h b_w$ samples stored at q bits each.

F. Receiver

Upon reception, the decoder obtains the stego index table and the codebook, extracts the first-layer payload from the seed blocks, and—using the same side-match rule together with the stored L -bit prefix—uniquely recovers each seed's original index; for residual blocks judged embeddable, it extracts L bits per block, then ERLE-decodes the indication stream to reconstruct the residual-region indication map of size $(I_h/b_h - 1) \times (I_w/b_w - 1)$, which enables verification and exact reproduction of the embedding state. The outputs are the concatenated secret bitstream (decodable to plaintext) and the recovered pre-embedding index table; in the logical-embedding baseline, the recovered table equals the input table.

IV. EXPERIMENTS

This section presents a detailed evaluation of the proposed method, covering its results and performance, and compares it with existing approaches. We first assess, across all test images, both security and embedding capacity (payload). We then examine model performance under different codebook sizes. Finally, we conduct a comparative analysis against alternative methods to contextualize the advantages and limitations of our approach.

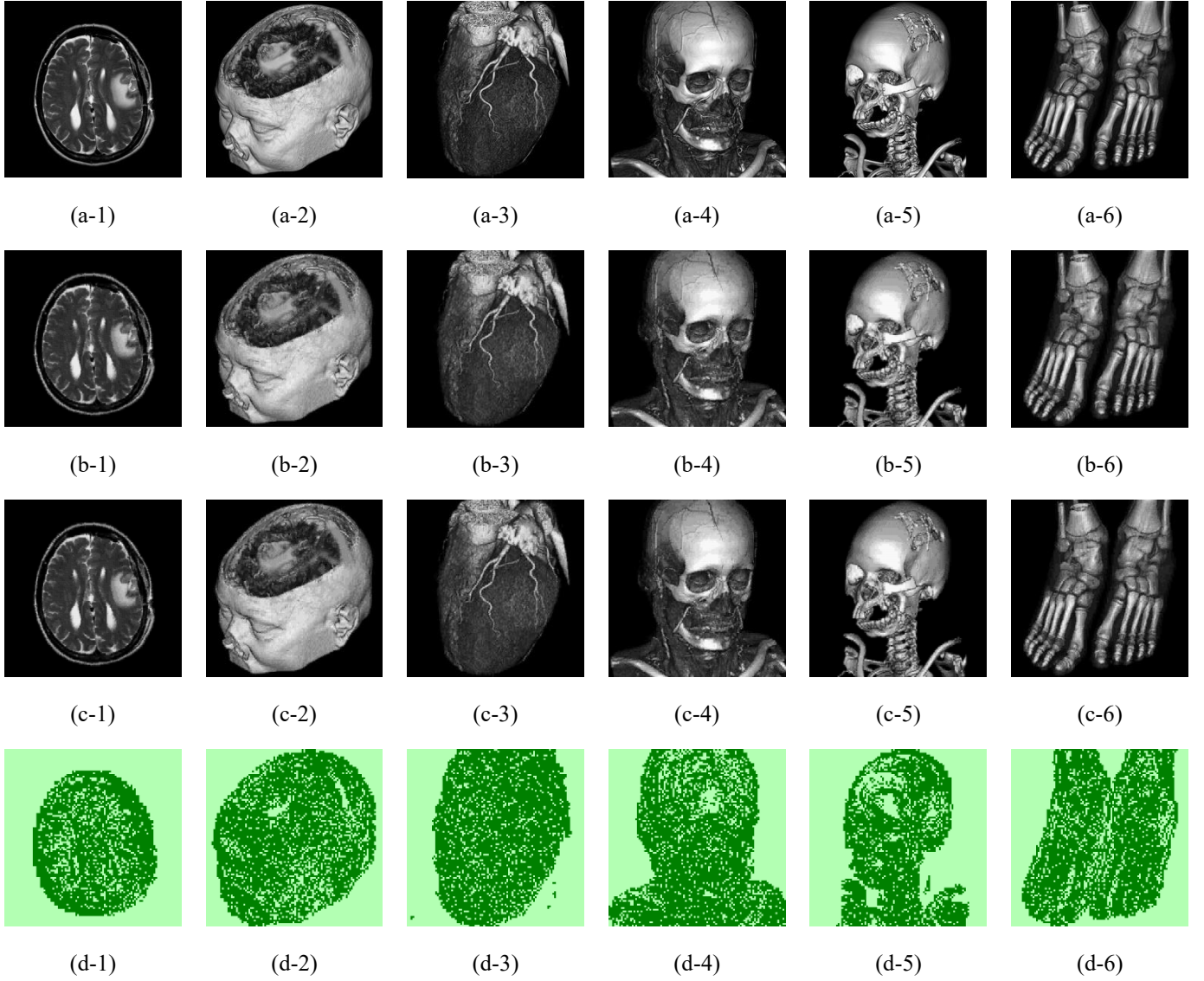


Fig. 4. Proposed method results (codebook size = 64). (a-1)–(a-6) show the original images; (b-1)–(b-6) depict the encrypted images; (c-1)–(c-6) present the decrypted images; and (d-1)–(d-6) illustrate the distribution of embeddable (data-hiding) blocks.

A. Experimental Setups

We evaluate the method on six widely used medical images BRAINIX, CEREBRIX, GOUDURIX, MANIX, PHENIX, and VIX in Fig. 4, each at 400×400 pixels; experiments run on a desktop with an AMD Ryzen 9 7900X3D 12-core CPU, an NVIDIA GeForce RTX 4070 Ti SUPER GPU, and 64 GB RAM under Windows 11 Home; and metrics include visual quality of VQ-coded images, embedding capacity across codebook sizes $K \in \{64, 128, 256, 512\}$, and lossless reconstruction performance as a function of K .

B. Performance Analysis

In our experiments, we first trained a codebook for each image and then applied a two-layer embedding strategy to

conceal the secret message within the VQ index table. This yielded encrypted images carrying hidden information, which we subsequently decoded to recover both the secret message and the decrypted images, ensuring that the pre-embedding index table and the post-decoding index table are identical—thereby demonstrating reversibility. We also evaluated the payload to assess the method’s performance. To further substantiate the security and fidelity of hiding information in the VQ index table, we conducted experiments on six widely used medical images. As shown in Fig. 4, panels (a-1)–(a-6) are the original images, (b-1)–(b-6) are the encrypted images, and (c-1)–(c-6) are the decrypted images. The results indicate that, after embedding and subsequent recovery, the images exhibit no perceptible degradation, which is desirable for medical

imagery that requires high-fidelity covert embedding.

To more precisely quantify imperceptibility (security), we compare the PSNR of the original and encrypted images and stratify the results by codebook size. The PSNR is defined as

$$PSNR(dB) = 10 \log_{10} \left(\frac{(MAX_I)^2}{MSE} \right) \quad (6)$$

$$MSE = \frac{1}{WH} \sum_{x=1}^W \sum_{y=1}^H (I(x, y) - \hat{I}(x, y))^2 \quad (7)$$

where I and $\hat{I}$ denote the original and encrypted (or reconstructed) images of size WH , and MAX_I is the peak

possible pixel value (255 for 8-bit images).

As reported in Table I, the PSNR increases monotonically with the codebook size $K \in \{64, 128, 256, 512\}$. This positive correlation arises because larger K provides more candidate codewords per block, thereby reducing quantization error and yielding reconstructions closer to the originals—hence higher PSNR. This trend corroborates the method’s imperceptibility: embedding via the VQ index table preserves visual fidelity even as payload is carried.

TABLE I
EVALUATION OF PSNR

Images	PSNR(dB) in K = 64	Images	PSNR(dB) in K = 128
BRAINIX	28.0591	BRAINIX	29.1772
CEREBRIX	26.8976	CEREBRIX	27.8155
GOUDURIX	25.9398	GOUDURIX	26.7423
MANIX	27.0800	MANIX	27.9218
PHENIX	24.3219	PHENIX	25.1999
VIX	28.0417	VIX	29.0719
Images	PSNR(dB) in K = 256	Images	PSNR(dB) in K = 512
BRAINIX	30.2202	BRAINIX	31.2299
CEREBRIX	28.6353	CEREBRIX	29.4587
GOUDURIX	27.5679	GOUDURIX	28.3449
MANIX	28.7858	MANIX	29.6901
PHENIX	26.0142	PHENIX	26.8569
VIX	29.9776	VIX	30.8909

We further analyze the payload capacity by examining the embedding capacity under different codebook sizes K , as summarized in Table II. The results show that the embedding capacity decreases slightly as K increases, indicating an inverse relationship. Intuitively, when the codebook grows larger, it introduces many mutually dissimilar codewords; the candidate “bucket” for each block becomes less homogeneous, which in turn lowers the probability that the original codeword remains the argmin, thereby reducing the effective payload.

Taken together with the PSNR results, this trade-off is acceptable. Increasing K improves reconstruction fidelity (higher PSNR) at the expense of a modest reduction in payload, whereas decreasing K boosts payload at some cost in image quality. Practitioners can therefore choose K according to

application requirements, balancing imperceptibility and capacity for the target use case.

C. Comparison with Related Works

In Table III, we compare embedding capacity against prior work. For a fair benchmark, we report results using two codebook sizes, $K = 64$ and $K = 256$, and evaluate on common grayscale test images (*Airplane*, *Boat*, *Peppers*, *Goldhill*, *Tiffany*). The results show that our method achieves substantially higher capacity in nearly all cases when $K = 64$. Even in a more quality-oriented setting ($K = 256$), our approach still outperforms competing methods in most scenarios. This demonstrates that the proposed scheme maintains a favorable balance between capacity and fidelity across typical evaluation settings.

TABLE II
EVALUATION OF EMBEDDING CAPACITY

Images	EC in K = 64	Images	EC in K = 128
BRAINIX	25508	BRAINIX	23349
CEREBRIX	19815	CEREBRIX	16450
GOUDURIX	19380	GOUDURIX	16021
MANIX	19091	MANIX	15480
PHENIX	23798	PHENIX	22374
VIX	20798	VIX	16420
Images	EC in K = 256	Images	EC in K = 512
BRAINIX	21536	BRAINIX	20182
CEREBRIX	11633	CEREBRIX	9852
GOUDURIX	13518	GOUDURIX	12122
MANIX	12354	MANIX	10724
PHENIX	19911	PHENIX	18242
VIX	13720	VIX	13818

TABLE III
EVALUATION OF EMBEDDING CAPACITY WITH SOC-BASED SCHEMES

Images	[5]	[21]	Ours (K=256)	Ours (K=64)
Airplane	21,838	13,664	20630	27660
Boat	20,696	13,762	13474	17852
Peppers	22,013	13,729	23119	28531
Goldhill	---	---	10544	19037
tiffany	---	---	16452	27644

D. Comparison with Related Works

In Table III, we compare embedding capacity against prior work. For a fair benchmark, we report results using two codebook sizes, $K = 64$ and $K = 256$, and evaluate on common grayscale test images (*Airplane*, *Boat*, *Peppers*, *Goldhill*, *Tiffany*). The results show that our method achieves substantially higher capacity in nearly all cases when $K = 64$. Even in a more quality-oriented setting ($K = 256$), our approach still outperforms competing methods in most scenarios. This demonstrates that the proposed scheme maintains a favorable balance between capacity and fidelity across typical evaluation settings.

V. CONCLUSION

We introduced a reversible two-layer data-hiding scheme for VQ-compressed images that operates on the index table rather than pixels. The design utilizes seed/residual block assessment via side-match evaluation, a 4-bit prefix to ensure unique index recovery for embeddable blocks, and ERLE to compress the residual-region indication map, all within a fixed 32-bit index budget per 4 blocks—preserving codec compatibility and exact reversibility.

Experiments on six medical images show that visual fidelity remains high: the PSNR increases with codebook size K due to reduced quantization error, whereas the embedding rate declines slightly as the candidate sets become less homogeneous. The capacity–fidelity trade-off is thus

controllable: smaller K yields higher payload with modest quality cost; larger K improves imperceptibility with minor capacity loss. Payload is reported as effective capacity (EC), defined as the total embedded bits from seed and residual blocks minus the ERLE-compressed indication-bit overhead.

Comparisons of standard grayscale images indicate substantial gains at $K = 64$ and competitive or superior results at $K = 256$, evidencing a strong balance among imperceptibility, capacity, and reversibility for clinical use. Future work will explore adaptive seed/residual rules and candidate sets to raise “found” rates at larger K , learned or content-aware indication coding beyond ERLE, robustness under compression/transcoding and adversarial analysis, as well as integration with alternative quantizers and hybrid codecs.

Data Availability

The data used to support the findings of this study are included within the article.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Author Contributions

Conceptualization, F.-Y.C., Y.L. and C.-C.C. (Chin-Chen Chang); methodology, F.-Y.C., Y.L. and C.-C.C. (Chin-Chen Chang); software, F.-Y.C.; validation, F.-Y.C.; formal analysis, C.-C.C. (Ching-Chun Chang) and C.-C.C. (Chin-Chen Chang); investigation, F.-Y.C. and Y.L.; resources, C.-C.C. (Chin-Chen Chang); data curation, F.-Y.C.; writing—original draft preparation, F.-Y.C.; writing—review and editing, Y.L. and C.-C.C. (Chin-Chen Chang); visualization, F.-Y.C. and Y.L.; supervision, C.-C.C. (Ching-Chun Chang) and C.-C.C. (Chin-Chen Chang); project administration, C.-C.C. (Ching-Chun Chang) and C.-C.C. (Chin-Chen Chang). All authors have read and agreed to the published version of the manuscript.

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